

When Does Multiview Fusion Help Entity Resolution? A Fine-Grid, Cross-Regime Study of Clustering Paradigm, Fusion Order, and Signal Complementarity

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Abstract

Multiview entity resolution (ER) combines partial views of a record — name, address, title, venue — to decide which records refer to the same entity, yet practitioners lack guidance on *when* fusion helps. We organize the design space as three factors: the **clustering paradigm** (pair-wise/metric vs. graph/topological), the **fusion order** (early fuse-then-cluster vs. late cluster-then-consensus), and the **ER regime** (deduplication vs. record linkage), and evaluate them over a fine grid of *configurations* — the unit of analysis — on five real and synthetic corpora spanning both regimes. Three results emerge. First, a **coarse grid can invert the scientific conclusion**: an equal-weight grid labels the CORA bibliographic corpus “fusion hurts,” but a fine weight-swept grid shows fusion helps (title+venue, not authors), a reversal we stress-test with a **matched-budget CGI audit** that separates genuine gains from search-space inflation. Second, the **pairwise paradigm strongly dominates the graph paradigm in both regimes**: graph cells repeatedly collapse through giant-component percolation, while fusion order is negligible in deduplication but decisive under linkage (early $\gg$ late). Third, whether and how much fusion helps is graded by **signal complementarity** — a three-type taxonomy (redundant-dominant, weak, strong) that classifies all six view sets and rises monotonically with the fusion benefit. Finally, a supervised **logistic-regression combiner** achieves near-perfect ranking ($AUC \approx 1.0$) but only matches fixed-weight fusion for clustering accuracy, the operative lever being decision-threshold calibration under extreme class imbalance rather than model complexity. Fusion benefit in ER is thus conditional, complementarity-graded, and largely paradigm- and calibration-bound — not a property a single “method” wins outright. Every number cites a result file and a `configuration_id`.

1 Introduction

Entity resolution (ER) — deciding which records refer to the same real-world entity — underpins data integration, deduplication, and record linkage [3–5]. Records are usually *multiview*: a person exposes name, address, and locality; a citation exposes title, authors, and venue. A long line of work fuses such views, yet practitioners lack a simple answer to a basic question: *given a new dataset, will multiview fusion help, and how should I combine the views?*

We make three contributions. (i) We recast the design space as a three-factor framework (§2) and adopt the *configuration* — a fully specified pipeline instance — as the unit of analysis. (ii) We show that the answer to “does fusion help?” is sensitive to grid resolution: a coarse grid produces a *false* negative conclusion on CORA that a fine grid overturns (§5), and we introduce a matched-budget audit so the correction is not itself a search-space artifact. (iii) Across deduplication (§6) and

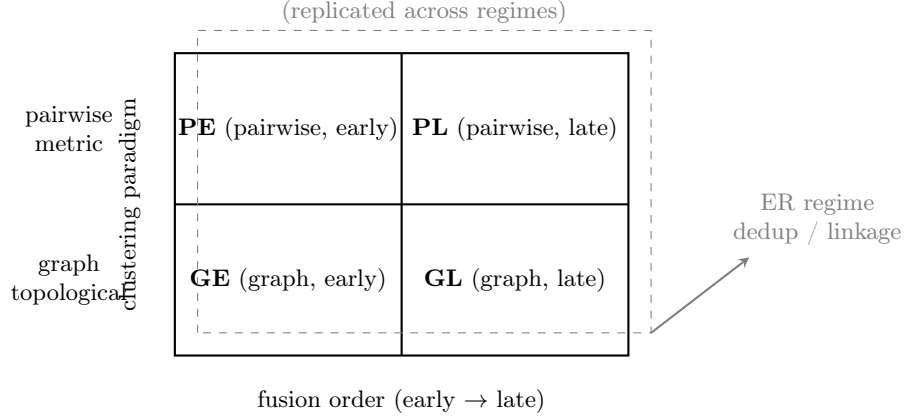


Figure 1: The three-factor framework: clustering paradigm $\times$ fusion order (the four method cells PE/PL/GE/GL), replicated across the deduplication and record-linkage regimes.

Table 1: The four method cells: clustering paradigm $\times$ fusion order. The logistic ML combiner (§8) is a supervised decision-rule *variant* of the pairwise cells, not a new axis.

	early (fuse $\rightarrow$ cluster)	late (cluster $\rightarrow$ consensus)
pairwise / metric	PE: <code>sim_combine</code> $S = \sum_v w_v S_v$, threshold, connected components	PL: threshold-cluster each view, co-membership consensus at α
graph / topological	GE: fuse graphs (multiplex / SNF), community detection	GL: community-detect each view graph, consensus at α

record linkage (§7) we find the pairwise paradigm strongly dominates the graph paradigm, and we propose a three-type complementarity taxonomy (§9) that predicts the sign and magnitude of the fusion benefit. A supervised logistic combiner (§8) matches but does not beat fixed-weight fusion, which bounds how much classical ML adds here.

2 A Three-Factor Framework

We organize multiview ER along three factors (Figure 1, Table 1):

- **Clustering paradigm** — are views combined and resolved by an absolute similarity cutoff (*pairwise/metric*) or by graph topology such as community detection and diffusion (*graph/topological*)? A similarity matrix S_v is the weighted adjacency of a graph G_v , so the axis is not the data structure but the edge-selection operation: an absolute cutoff versus topology-driven clustering with no cutoff.
- **Fusion order** — *early* fuses all views into one object and clusters once; *late* clusters each view and reconciles by co-membership consensus.
- **ER regime** — *deduplication* (within one source, all-pairs) vs. *record linkage* (two sources, bipartite pairs). The regime is a property of the task and data, *not* a method knob; it is realized by choosing datasets, so the four method cells are replicated across regimes.

The pairwise cells combine similarities with `sim_combine`, $S(i, j) = \sum_v w_v S_v(i, j)$, then threshold and take connected components; the graph cells fuse per-view graphs by multiplex supra-adjacency [9] or similarity network fusion [12] and cluster with Louvain [1].

Table 2: Dataset summary. Two ER regimes; five corpora. PC = pair completeness (recall ceiling of blocking).

Dataset	Regime	n	Views	Pairing	PC
CORA	dedup	1,879	title, authors (, venue)	all-pairs	—
FEBRL1	dedup	1,000	name, address, geo	all-pairs	—
DBLP-ACM	linkage	4,910	title, authors, venue	SN(title, $w=20$)	0.985
Restaurant	linkage	864	name, address, type	full bipartite	1.000
NCVR-2party	linkage	4,000	name, geo	SN(surname, $w=40$)	0.868

Table 3: Matched-budget CGI audit. Raw CGI>1 for all six view sets, but the multiview search explores far more configurations than the single-view search. Re-drawing the multiview search at the single-view budget, $P(\text{win})$ is the fraction of equal-budget searches that still beat the best single view. Restaurant and CORA-2view are *fragile*; the rest are robust.

View set	Regime	raw CGI	$P(\text{win})$	median matched	verdict
CORA-2view	dedup	1.022	0.957	0.878	fragile
CORA-3view	dedup	1.059	0.997	0.910	robust
FEBRL1	dedup	1.162	1.000	0.973	robust
Restaurant	linkage	1.082	0.376	0.819	fragile
DBLP-ACM	linkage	1.053	0.923	0.893	robust
NCVR-2party	linkage	1.208	0.949	0.879	robust

3 Dataset Regimes

We use five corpora across the two regimes (Table 2). **Deduplication:** CORA (bibliographic; views title, authors, and venue = coalesce(journal, booktitle)) and FEBRL1 (synthetic person records; views name, address, geo) [2]. **Record linkage:** DBLP-ACM and Restaurant (Fodors-Zagat) from the standard benchmark suite [8], and NCVR-2party, a two-source subset of North Carolina voter registrations [10]. Because full-bipartite similarity is infeasible at scale, the two large linkage corpora use sorted-neighborhood blocking [11]; we report pair completeness (PC), the recall ceiling it imposes. Near-unique identifiers (FEBRL1 `soc_sec_id/date_of_birth`) are excluded as leakage.

4 Configuration Design

The **configuration** is the unit of analysis: a fully specified pipeline instance (paradigm, order, views, per-view similarity, weighting, normalization, threshold, and cluster method), each producing one evaluated result row with a stable `configuration_id`. For the pairwise-early cell we sweep a fine grid: threshold τ in fine steps, the weight simplex over views, named weightings (equal, variance, Fellegi-Sunter [5]), and normalization. We score partitions by the adjusted Rand index [7] (primary), with NMI, VI, FMI, and pairwise F1/precision/recall.

The Complementary Gain Index and its audit. For per-view best ARI a_v and best fused ARI $a_\star$, the *Complementary Gain Index* is $\text{CGI} = a_\star / \max_v a_v$ (fusion helps iff $\text{CGI} > 1$). Because $a_\star$ is a maximum over a much larger search than a_v , we pair every CGI with a **matched-budget audit**: we re-draw the multiview search at the single-view budget and report the fraction of equal-budget searches that still beat the best single view (Table 3). This distinguishes a genuine gain from search-space inflation.

Table 4: Fine-grid correction on CORA. A coarse equal-weight grid places the best fusion *below* title-only (“fusion hurts”); a fine weight-swept grid places it *above* (“fusion helps”). The best 3-view configuration sets **authors**= 0: venue, not authors, is the useful complement.

CORA view set	coarse best	fine best	title-only	CGI (fine)
title+authors (2-view)	0.7025	0.8874	0.8681	1.022
title+authors+venue (3-view)	0.7069	0.9189	0.8681	1.059
<i>3-view contribution (fine grid, best ARI):</i>				
title 0.868 authors 0.622 venue 0.444	(single views)			
title+authors 0.887 title+venue 0.919 title+authors+venue 0.916				

Table 5: Deduplication results. Best pairwise-early (PE) configuration per view set (fine grid), with the best graph cell inherited from the coarse 2×2 study[†]. CGI = best-multiview / best-single-view.

View set	single	PE	NMI	VI	FMI	graph [†]	CGI	Type
CORA-2view	0.868	0.8874	0.951	0.391	0.892	0.629	1.022	I
CORA-3view	0.868	0.9189	0.914	0.724	0.922	0.629	1.059	II
FEBRL1	0.849	0.9859	0.999	0.017	0.986	0.735	1.162	III

5 Fine-Grid Correction

A coarse, equal-weight grid on CORA places the best fusion *below* the best single view and would support a “fusion hurts” conclusion. A fine, weight-swept grid overturns this: the best two-view fusion reaches ARI 0.8874 and the best three-view fusion 0.9189, both *above* title-only 0.8681 (Table 4). The correction is mechanistic, not cosmetic: the three-view optimum sets **authors**= 0, so the useful complement is *venue*, not authors — title+venue reaches 0.919 while adding authors on top (title+authors+venue = 0.916) slightly *hurts*. This is a concrete instance of a general hazard: **coarse hyperparameter grids can produce false scientific conclusions**. The response surface is a broad plateau (tens of configurations within 0.01–0.02 of the optimum across a threshold band and both viable normalizations), so the corrected optimum is stable, not a knife-edge.

6 Deduplication Results

Table 5 summarizes the deduplication regime. The best pairwise-early configuration beats the best single view on all three view sets, but the matched-budget audit (Table 3) shows the *magnitude* and *robustness* vary sharply: FEBRL1 is decisive (matched-budget win probability 1.000, median gain +0.125), CORA-3view robust (0.997, +0.042), and CORA-2view fragile (0.957, +0.010 — roughly half its raw margin is search-space inflation). Graph cells, inherited from the coarse 2×2 study[†], trail the pairwise cells on both corpora.

7 Record-Linkage Results

Table 6 reports all four cells under linkage. Two findings stand out. First, the **pairwise paradigm strongly dominates the graph paradigm**: pairwise cells reach ARI 0.88–0.92 while graph cells (GE, GL) collapse to ≤ 0.22 — with high recall but near-zero precision, the signature of **giant-component percolation** on the bipartite candidate graph (largest predicted clusters of 143–245 records against a true size of ≈ 2). This mirrors the graph collapse observed in deduplication and

Table 6: Record-linkage results (best ARI per cell). Pairwise cells dominate; graph cells (GE, GL) collapse via giant-component percolation (high recall, near-zero precision). Early (=PE) beats late (=PL) throughout.

Dataset	PE	PL	GE	GL	PE prec.	PE rec.	Type
DBLP-ACM	0.9039	0.834	0.019	0.221	0.866	0.946	II
Restaurant	0.9224	0.566	0.006	0.062	0.927	0.918	I
NCVR-2party	0.8806	0.656	0.002	0.002	0.961	0.813	II→III

Table 7: Logistic ML combiner (E2a) vs. fixed-weight PE and best single view, on entity-disjoint *test* sets[‡]. AUC is near-perfect everywhere, but logistic only matches fixed-weight fusion for clustering ARI (wins DBLP, ties NCVR, loses Restaurant). Optimal thresholds are 0.90–0.99, never 0.5 — a class-imbalance calibration effect.

Dataset	logistic	PE	single	AUC	best thr	learned coeffs (sign)
DBLP-ACM	0.9763	0.9384	0.9339	0.999	0.90	title+, authors+, venue+
Restaurant	0.9577	0.9722	0.9117	0.999	0.99	name+, address+, type −
NCVR-2party	0.9102	0.9044	0.8266	1.000	0.95	name+, geo+

is therefore regime-invariant. Second, **fusion order matters under linkage**: early (PE) beats late (PL) on every dataset (by +0.07 to +0.36 ARI), because co-membership consensus loses recall when the two sources disagree across views. All three linkage corpora have CGI>1, robust under the matched budget for DBLP-ACM and NCVR but *fragile* for Restaurant ($P = 0.376$; Table 3).

8 A Supervised Logistic Combiner

We test whether a supervised model beats fixed-weight fusion by fitting logistic regression [6] on per-view similarity features, using an **entity-disjoint** 70/30 split so no entity appears in both train and test, and balanced class weights for the extreme imbalance (positive rates 0.0009–0.06). Predicted probabilities are thresholded and clustered by connected components, then compared on the *same* held-out test set[‡] against the fixed-weight pairwise-early baseline and the best single view (Table 7).

Logistic regression attains near-perfect ranking ($AUC \approx 0.999$ –1.000) but only *matches* fixed-weight fusion for clustering ARI: it wins on DBLP-ACM (+0.038), ties on NCVR (+0.006), and loses on Restaurant (−0.015). The learned coefficients confirm the complementarity structure — restaurant **type** receives a *negative* coefficient (same-type restaurants are common non-matches), NCVR **geo** a large positive one — but the discriminative task is already saturated. The operative lever is calibration: the optimal decision threshold is 0.90–0.99, never 0.5, a direct consequence of class imbalance. Because AUC is already at ceiling and the residual bottleneck is threshold calibration and blocking recall rather than model capacity, more flexible learners (random forests, gradient boosting) are not warranted on this evidence, and we do not pursue them.

9 A Complementarity Taxonomy

The results across both regimes are organized cleanly by **signal complementarity** rather than by a binary label (Figure 2, Table 8). We distinguish three types: **Type I** redundant-dominant (one view dominates, others duplicate it; fusion neutral or fragile), **Type II** weak complemen-

Table 8: The three-type complementarity taxonomy classifies all six view sets. The fusion gain (best multiview – best single view, matched-budget) rises monotonically I→II→III.

Type	Definition	Members (matched-budget gain)	Behavior
I	redundant-dominant	CORA-2view (+0.010), Restaurant (fragile)	fusion $\approx$ neutral / fragile
II	weak complementarity	CORA-3view (+0.042), DBLP-ACM (+0.035)	small, robust gain
II→III	moderate	NCVR-2party (+0.150)	large, robust gain; one dominant view
III	strong complementarity	FEBRL1 (+0.125)	large gain; no dominant view

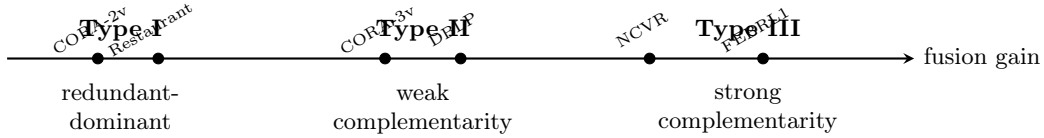


Figure 2: The complementarity taxonomy. Matched-budget fusion gain increases from Type I (redundant-dominant) through Type II (weak) to Type III (strong). All six view sets place without forcing.

tarity (a dominant view plus a modestly complementary one; small robust gain), and **Type III** strong complementarity (no dominant view; large gain). The matched-budget fusion gain rises monotonically across the types, and every view set classifies without forcing: CORA-2view and Restaurant are Type I, CORA-3view and DBLP-ACM Type II, NCVR-2party Type II leaning III, and FEBRL1 the one clear Type III. The same corpus can change type with its view set (CORA is I with title+authors, II with title+venue), which makes the view set — not the dataset — the honest unit of geometric analysis.

10 Threats to Validity

- **No real Type III linkage dataset.** Strong complementarity rests largely on the single synthetic FEBRL1 corpus; NCVR is only Type II leaning III. The strong-complementarity end of the taxonomy is under-sampled in the linkage regime.
- **Blocking recall ceiling.** NCVR uses sorted-neighborhood blocking with $PC = 0.868$, so $\approx 13\%$ of true matches are unrecoverable; its absolute ARIs are correspondingly depressed and cross-dataset absolute levels must be read with PC in mind.
- **Logistic only.** The supervised study covers the linear family; “ML does not beat fixed fusion” is established for logistic regression and not for tree ensembles, which we did not run.
- **Deduplication ML not run.** The supervised combiner was evaluated on the linkage corpora only; the deduplication ML comparison is deferred.
- **Matched-budget CGI is an audit, not an estimator.** It resamples to an equal budget but is not a complexity-corrected, inference-grade quantity with confidence intervals or nested selection.
- **Inherited graph baselines.** [†]Deduplication graph-cell values are inherited from the earlier coarse 2×2 study because the fine-grid stage reran only the pairwise-early cell. [‡]The logistic results are entity-disjoint test-set comparisons and are not directly comparable to the full-record best-pairwise values in Table 5–6.

11 Conclusion

Reframing multiview ER as clustering paradigm $\times$ fusion order $\times$ ER regime, and adopting the configuration as the unit of analysis, yields a cleaner and more honest account than method-vs-method comparison. Across deduplication and record linkage, the pairwise/metric paradigm strongly outperforms the graph/topological paradigm, which frequently collapses through giant-component percolation; a coarse grid can invert the fusion verdict, so the matched-budget audit is essential; and whether fusion helps is graded by a three-type complementarity taxonomy rather than a binary label. A classical logistic combiner matches but does not beat fixed-weight fusion, locating the remaining leverage in threshold calibration and candidate generation, not model complexity. The principal open step is a real strong-complementarity (Type III) linkage corpus to complete the taxonomy across regimes. Every result is tied to a reproducible `configuration_id`.

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